horticulturae

MDPI

Review

Sustainable Agriculture Through Compost Tea: Production, Application, and Impact on Horticultural Crops

Emanuela Campana, Michele Ciriello *, Matteo Lentini, Youssef Rouphael and Stefania De Pascale *

Department of Agricultural Sciences, University of Naples Federico II, 80055 Portici, Italy; emanuela.campana@unina.it (E.C.); matteo.lentini@unina.it (M.L.); youssef.rouphael@unina.it (Y.R.)
* Correspondence: michele.ciriello@unina.it (M.C.); depascal@unina.it (S.D.P.)

Abstract: As part of the European Green Deal, the Farm to Fork strategy was introduced with the idea that environment, agriculture and food are interconnected topics. Reducing the use of synthetic fertilizers by 20% before 2030 through the adoption of circular economy principles is one of the goals to be achieved. There are several bioproducts that can be obtained from the valorization of agro-industrial wastes used to increase crop yields under low-fertilizer applications. However, the aim of this review is to describe production methods and the use of compost tea on horticultural crops to understand its real potential in providing plant growth support. The effects of compost tea on crops can vary widely depending on the waste material used, compost quality, compost tea production process and parameters, and the interaction between horticultural species and compost tea application dose. Therefore, because of this heterogeneity, it is possible that we would achieve real, positive impacts on the environment and horticultural production if there were more collaboration between the research sector and private farms. This collaboration would allow the development of protocols for compost tea production and customized use according to real farm needs. This would reduce both the costs associated with the disposal of waste produced on the farm and reduce the costs associated with the supply of synthetic fertilizers. The adoption of on-farm guidelines for compost tea use would achieve a balanced trade-off between agricultural productivity and environmental sustainability. The literature review shows that the most-used dilution ratios, regardless of the type of starting compost, range from 1:5 to 1:10 compost–water (v/v). Although a complete understanding of the biostimulatory mechanisms activated by compost tea is lacking, the application of this bioproduct would improve the physiological and productive performance of many horticultural species of interest, especially under suboptimal conditions such as organic production.

Keywords: European green deal; synthetic fertilizers; sustainable tools; vegetables; compost; circular economy; physiological responses

check for updates

Academic Editor: Luigi De Bellis

Received: 1 March 2025
Revised: 10 April 2025
Accepted: 14 April 2025
Published: 18 April 2025

Citation: Campana, E.; Ciriello, M.; Lentini, M.; Rouphael, Y.; De Pascale, S. Sustainable Agriculture Through Compost Tea: Production, Application, and Impact on Horticultural Crops. *Horticulturae* **2025**, *11*, 433. https://doi.org/10.3390/horticulturae11040433

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Since the 1960s, with the aim of adequately responding to growing food demand, farmers have used over-optimal doses of fertilizers to increase crop yields [1]. However, this strategy has not turned out to be a successful one. The main reason concerns the low-crop-fertilizer-use efficiency (FUE). For instance, from the total nitrogen fertilizer applied to grow the staple cereal crops globally (rice, wheat, and corn), about 45% is leached or lost to the environment [2]. In light of this, an increase in fertilizer inputs, rather than resulting in increased yield, would result in an unsustainable increase in

production costs for the farmer [3]. As if this were not enough, the inefficient use of fertilizers also negatively impacts the environment, promoting the degradation of the chemical and physical characteristics of soils, the eutrophication of water, and the emission of greenhouse gases into the atmosphere [4]. It is also important to consider that very often farmers not only use low-quality fertilizers (with reduced nutrient supply and high levels of impurities) but also incorrectly choose how and when to apply synthetic chemical fertilizers [3]. The combination of these issues has prompted the European Union to adopt policies aimed at reducing the environmental impact produced by the agricultural sector. Within the proposed European Green Deal, there is the Farm to Fork strategy that includes some of the targets to be achieved by the agricultural sector by 2030. These include reducing synthetic fertilizers by at least 20% [5]. All that remains to be asked at this point is how farmers can reduce the input of synthetic fertilizers without compromising crop yields, in a global context where agricultural soils are mostly degraded, and their physical and chemical characteristics that were already compromised long ago. In this context, enhancing agro-industrial wastes into bioproducts that can provide or stimulate nutrient uptake by crops could be a promising strategy to pursue. The agricultural sector produces high amounts of organic and inorganic waste, many of which are rich in nutrients such as nitrates and phosphates. Of these, horticultural waste accounts for the largest slice of waste produced compared to all other foods [6], which is even expected to increase due to the intensification of agricultural activities due to the need to feed the growing global population [6]. Valorization of waste not only allows for reducing the potential risk of groundwater and soil pollution, but also avoids the costs derived from recovery, transportation, and landfill disposal [7]. Among the various processes of waste valorization, our focus is on transformation processes that are suitable for reducing dependence on synthetic fertilizers according to the principles of the circular economy. The transformation of agricultural waste through the process of composting is one of the most effective solutions to improve soil fertility [8]. Through the composting process, microbial degradation of organic waste is achieved in the presence of oxygen. This results in compost, which is a stable and mature material that is rich in nutrients and humic substances divided into three fractions: fulvic acids, humic acids and humins [9]. Compost is often used as a biofertilizer or soil conditioner, capable of providing "slow-release" nutrients for crop growth while increasing both microbial populations and soil water retention [10]. In addition to humic and fulvic acids, compost tea (CT) can be extracted from compost.

Compost tea is an aqueous extract of compost containing soluble nutrients and bioactive molecules extracted from compost and different microorganisms than those contained in the starting compost. Studies on CT have been conducted since the 1980s, offering a wealth of information on its potential uses to improve crop resistance to biotic stresses. Despite the heterogeneity of its characteristics and mode of action, the scientific literature is full of articles highlighting the crucial action of CT microorganisms in suppressing fungal and bacterial diseases in plants. Compost tea microorganisms can both produce compounds that suppress the growth of antagonistic fungi and bacteria and also stimulate plants to trigger their own defense mechanisms [11–13]. However, the biostimulatory action of CTs in supporting and improving crop growth has yet to be understood. Unlike compost, which releases nutrients gradually, CT would satisfy the needs of crops at specific phenological stages (transplanting, flowering, fruit set) where the need to provide nutrients is quite urgent [14]. Among its other advantages, compost tea can be used as a nutrient solution to be integrated into bioponic systems. These systems, also called "biological hydroponic systems", contrary to conventional hydroponic systems, deliver nutrients to crops through organic solutions [15].

The economic viability of producing CT is related to the type of waste material used for compost production, the composting method and the method of CT production. With regard to the waste material used, economic viability depends on its availability and abundance locally. In addition to this, a mix of different wastes must be used in order to obtain an ideal mix for transformation into compost. Therefore, the choice of waste to obtain the right mix must be well considered in order to minimize costs and produce good quality compost at the same time [16]. Based on previous studies, the average procurement cost for the solid fraction of pig slurry is USD 3.23 t^{-1}, for barley straw it is USD 8.62 t^{-1}, for forage by-product it is USD 6.46 t^{-1} and finally for prunings it is USD 7.54 t^{-1}. In an optimized composting process, the labor required (approximately 0.2 h per tonne of mixture) for the various operations to be performed during the composting process was estimated at USD 16.15 h^{-1}, while the sum of the other variable costs that are attributable to the composting phase is about USD 23.84 t^{-1} of mixture [16]. In addition, the costs for CT production depend on the production process: a CT produced by oxygenation will have higher production costs due to the electricity required to oxygenate the solution. Other factors influencing production costs concern the addition of additives and the time required to produce CT. According to a study by Matouk et al. [17], the operational costs for CT production were found to be significantly influenced by the addition of nutrients, the extraction temperature and the time required for CT production. Thus, although the use of CT avoids the costs associated with the disposal of agro-industrial waste, it is necessary to accurately choose the factors and time of CT production in order to reduce production costs. Pergola et al. [18] compared the economic and environmental aspects of strawberry cultivation in the regions of Basilicata and Campania (Italy) in eight different cultivation systems. The cultivation systems involving the use of CT were shown to be among the most profitable and able to cover the pollution costs produced. However, although the present work analyses cultivation systems through a life cycle analysis (LCA), there is a lack of information regarding the final destination of the solid compost residue after the CT extraction process. It might be interesting to explore the potential use of the solid compost residue as a soil conditioner in order to make the valorization of agro-industrial waste into CT even more convenient.

This review will be concerned with amalgamating the information in the scientific literature regarding the modes of action of CTs in improving the nutrient efficiency of crops, with a view to encourage the reduction of synthetic fertilizer inputs through the valorization of farm agricultural wastes. The outline of the process followed for the identification and selection of the articles analyzed in this review is shown in Figure 1.

Figure 1. Outline of the process followed for the identification and selection of the articles analyzed in this review.

Relevant studies were manually screened by reading the title, abstract and full text carefully in order to understand the correlation with the review topic. Figure 2 shows the trend of scientific papers on compost tea from 1996 to 2025. It is clear that scientific interest in this topic has been growing in recent years.

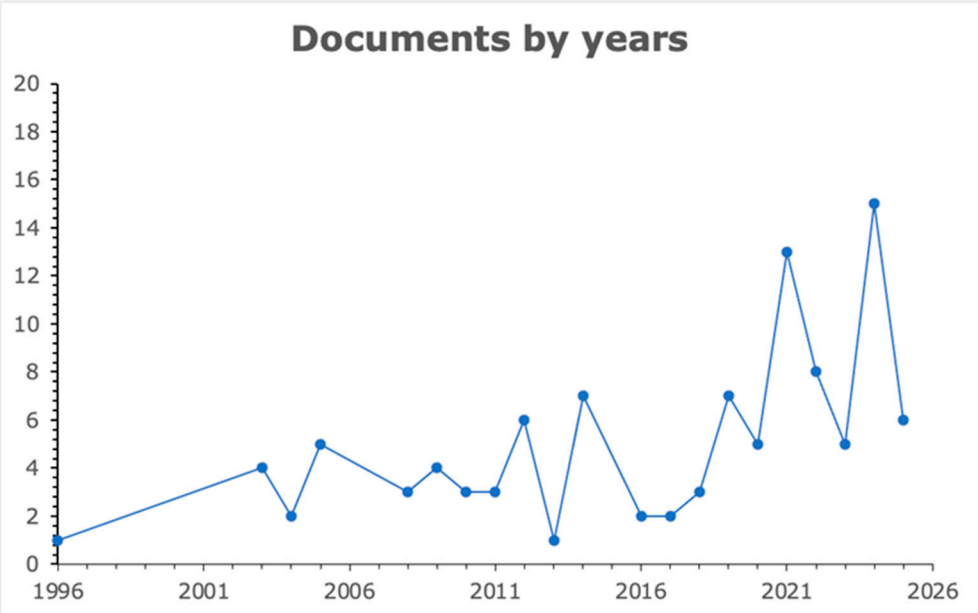

Figure 2. Trend of publications indexed on Scopus from 1996 to 2025 containing the term "compost tea". The search was limited to this exact keyword to avoid heterogeneous and ambiguous results due to the automatic separation of the terms "compost" and "tea" by the search engine.

In fact, publication peaks are observed in 2021 and 2024. Furthermore, Figure 3 shows the geographical distribution of articles published on compost tea over the same time period.

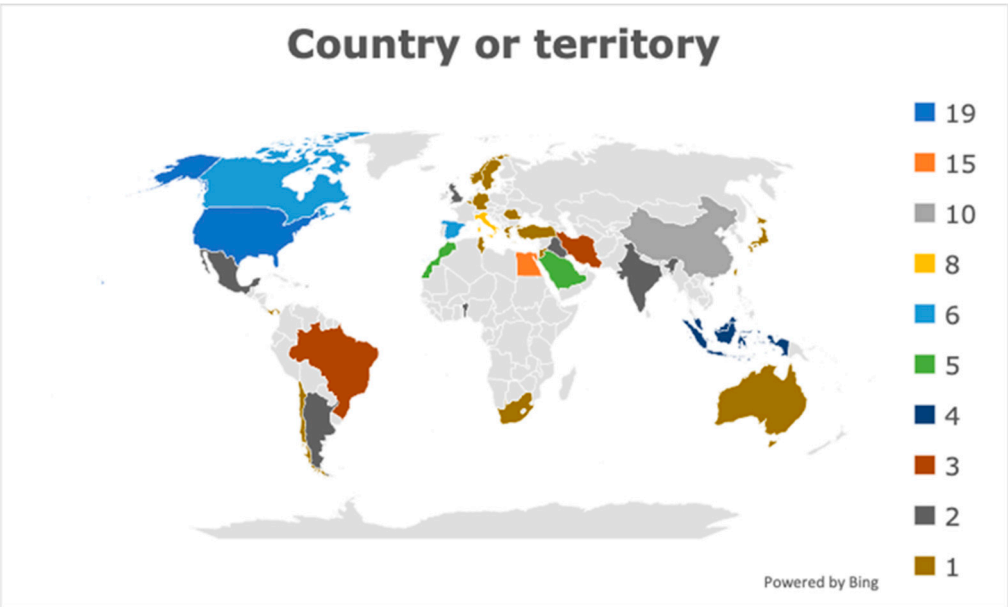

Figure 3. Geographical distribution of scientific articles on "compost tea" (1996–2025), ranked by number of publications: United States (19), Egypt (15), China (10), Italy (8), Canada (6), Spain (6), Morocco (5), Saudi Arabia (5), Indonesia (4), Malaysia (4), Brazil (3), Iran (3), Argentina (2), Benin (2), India (2), Iraq (2), Mexico (2), Puerto Rico (2), Trinidad and Tobago (2), United Kingdom (2), Australia (1),

Belgium (1), Chile (1), Germany (1), Greece (1), Japan (1), Jordan (1), Norway (1), Panama (1), Romania (1), South Africa (1), Sweden (1), Taiwan (1), Tunisia (1), and Turkey (1). The search on Scopus was limited to this exact keyword to avoid heterogeneous and ambiguous results due to the automatic separation of the terms "compost" and "tea" by the search engine.

For Figures 2 and 3, the search on Scopus was limited to the keyword "compost tea" and avoided ambiguous results due to the search engine's automatic separation of the terms "compost" and "tea".

2. Compost Tea: What It Is and How to Produce It

Elaine R. Ingham, a pioneer in the field of CT, in the fifth edition of her book "The compost tea brewing manual", defines this substance as "A brewed, water extract of compost" [19]. Compost tea is produced by a cold infusion process for a defined duration of time, which can facilitate both microbial proliferation and the extraction of soluble nutrients from the compost. However, there are two different methods of CT extraction, which differ in whether oxygen is introduced into the production system. Both extraction methods (aerated and non-aerated) involve infusing the compost (enclosed within a porous filter) in dechlorinated water for a period of time ranging from a few hours to several days [20]. The use of chlorinated water would hinder microbial proliferation during the infusion process, significantly reducing the bioactivity of the obtained CT [21–23]. Bioactivity is defined as the ability to induce a certain effect, response or reaction in plant tissues of crops. In general, bioactive effects depend on the bioactive substance, its bioavailability and its application dose [24]. The typical bioactive molecules present in CT are secondary metabolites extracted from compost and synthesized by microorganisms. These secondary metabolites are not required for normal plant growth processes, but play an important role in signaling pathways between plants, other organisms and the environment. At present, there is no clear relationship between structure and activity that can explain the mechanisms by which the bioactivity of certain molecules is expressed. The hypothesis described by Verrillo et al. [25] suggests that hormonal activity and the modulation of biochemical pathways are linked to the presence of lignin, aromatic compounds, polar compounds such as oligosaccharides, as well as amino acids and peptides observed in the CTs analyzed in this work. The conformational arrangement of the different molecules solubilized in the CT is also responsible for their bioactivity on the growth parameters of the crops [25]. The preparation of aerated compost teas involves the use of pumps to feed oxygen continuously or discontinuously. According to some authors, this method would allow for extracting a higher amount of soluble substances from the compost than can be extracted by the anaerobic fermentation process [9]. Ingham [19], in their book, affirms that as the concentration of oxygen in the solution decreases, a gradual reduction in nutrients is observed. The author argues that under conditions of anaerobiosis and reduced oxygen, nutrients are lost through volatilization. Un-aerated compost teas, on the other hand, involve incubating the compost in water for between 7 and 14 days [26]. These, unlike aerated compost teas, involve low energy costs and will consist mainly of anaerobic microorganisms.

In a study, it was observed that independently from the extraction method (aeration or fermentation) and the waste material used, the bacterial communities in the CTs were mainly composed of species belonging to the Proteobacteria and Bacteroidetes. In particular, *Proteobacteria* play a key role in the nitrogen cycle, promoting higher nitrate content in CTs. However, aerated CTs had a greater number and heterogeneity of species with high protease activity. In contrast, non-aerated CTs showed high cellulase activity and higher production of siderophores, metabolites involved in chelation processes [27]. Kim et al. [28] observed differences in microbial population density in CTs produced from

different composted materials. However, independently from the composted material used, the microbial communities of the aerated CTs were composed mainly of bacteria. The predominant bacterial species was *Bacillus*, followed by *Ochrobactrum*, *Spingomonas* and uncultured bacteria [28]. St. Martin et al. [29] also observed higher bacterial and fungal populations in aerated than in non-aerated CTs. This is probably because under anaerobic conditions, microorganisms have less energy available and therefore the growth rate is slowed down. In any case, regardless of the extraction method used, the microbial populations in CT may differ from those already present in the starting compost. This might be because the microorganisms in the compost, during the infusion period, transform the insoluble forms of nutrients into easily assimilated forms, promoting both the increase in mineral nutrients and humic acids in the CT, and the proliferation of a wide range of other microorganisms [9,21,30,31]. In this regard, it should be noted that Martínez-Yáñez et al. [32] compared the microbial populations of a mature home compost and its aqueous aerated extract. Microbial populations of the genera *Pantoea* (*Proteobacteria*), *Pseudomonas* (*Proteobacteria*) and *Pseudoxanthomonas* (*Pseudomonadota*) were not observed in the starting compost. Some microbial populations present in the starting compost were also not observed in the CT, especially bacteria of the genera *Lewinella* (*Bacteroidetes*) and *Thalassospira* (*Proteobacteria*). However, the information regarding metagenome differences between compost and CT is still very limited [33]. The microbial populations in the CT contribute to improving the fertility, water retention and ion exchange of soils through the production of secondary metabolites such as phytohormones, humic acids, chelated iron and organic acids [14]. However, anaerobic CTs generally have a higher nitrogen content. This could be due to the nitrification and denitrification reactions of aerobic microorganisms in aerated CTs, which would convert ammonia molecules into nitrate, nitrite and subsequently molecular nitrogen over time [29]. In contrast, no differences in P and K content were observed in aerated and non-aerated CTs, while higher Ca and Mg concentrations were observed in aerated CTs [29]. In another study, comparing the concentration of different chemical elements in aerated and fermented CTs, higher amounts of Na and Mo was observed in aerated CTs, while higher concentrations of K, Zn and Mn were observed in non-aerated CTs [34].

The numbers and types of final microbial populations present in the CT will also depend on whether or not additives and/or microorganisms are inoculated during the infusion process [19]. Very often, to promote microbial growth in the CT, nutritional additives such as molasses, yeast, fishmeal, whey or casein are added at the infusion stage [9]. However, the addition of such additives does not always guarantee a better final quality of CT [35]. In general, for both aerated and un-aerated CTs, longer infusion times result in an increase in soluble nutrients and antibiotics produced by microorganisms, which are useful for improving resistance to biotic stresses in crops [26]. Caution: if infusion times are extended too long, the microorganisms in the infusion will feed on the nutrients in the aqueous solution, decreasing the overall quality of the CT [26]. Compost tea can be applied either as is by foliar and root application or further diluted with dechlorinated water. Dilution and filtration are strongly recommended to preserve the operation of nozzles if CT is planned to be applied via irrigation systems [36].

3. From Compost-to-Compost Tea

The final characteristics of CT can be remarkably heterogeneous, depending on the different variables that are at play during the production process. Its properties, in fact, depend on the type of waste used for compost production, the final quality of the compost used for the CT extraction process, the compost/water dilution ratio, the production parameters (pH, production time, temperature, oxygenation), the addition or not of nu-

tritional additives and/or microbial consortia, and finally the CT/water dilution ratio during the application phase to the crop. However, it should be highlighted that due to the relationships existing between the different factors involved in the CT production process, it is impossible to distinguish exactly how and in which measure each of these may contribute in determining the final characteristics of the CT [27]. Making everything even more complex is the lack of standardization in the CT production method on an international and national level, which further increases the level of heterogeneity in the final products that can be obtained [14]. Figure 4 shows a schematic illustration of the CT production process and the factors affecting its quality.

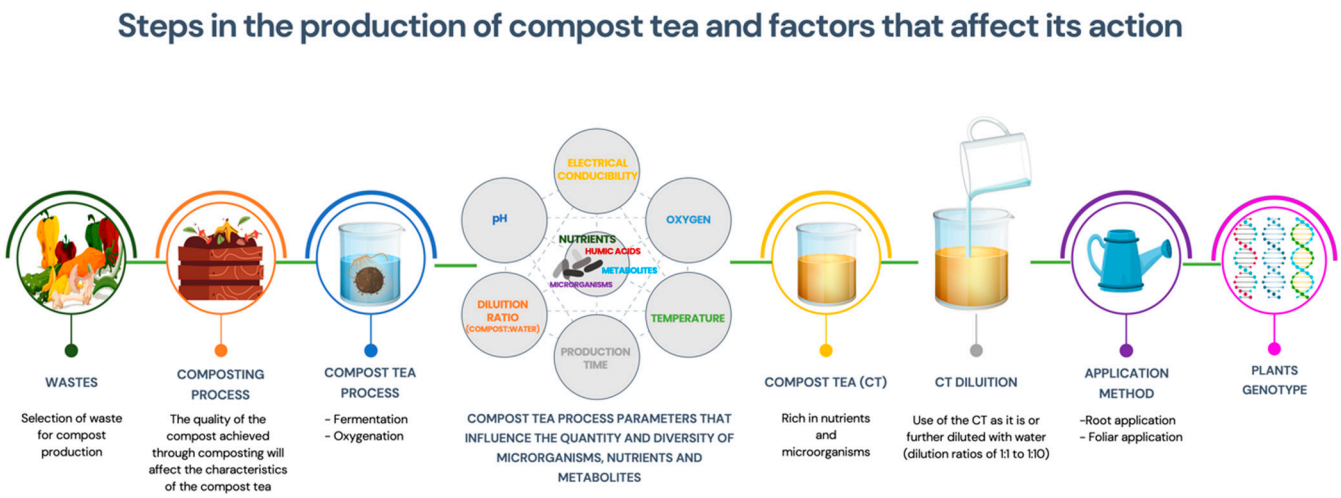

Figure 4. Outline of the compost tea brewing process and production parameters that affect its quality.

3.1. Waste Used for Compost Production

With the aim of enhancing the value of waste produced by the agricultural sector so as to reduce fertilizer inputs to crops and increase farmers' profitability, only waste from the agricultural sectors will be discussed. Compost can be obtained from different kinds of waste produced during the production phase of agricultural and livestock products. Waste generated from poultry and livestock farming includes manure, litter, feed and carcasses. Waste generated from agricultural and horticultural production includes crop residues such as sugarcane bagasse and corn straw, pruning residues, and plant residues such as rotten leaves, vegetables and fruits. These wastes have high potential for valorization, as they are still rich in nutrients such as nitrogen (N), phosphorus (P) and potassium (K) [6]. Ngakou et al. [37] observed higher N, P and K contents in CTs produced from cow dung compost compared to CTs produced from kitchen waste. St. Martin et al. [29], comparing the characteristics of CTs produced from banana leaf waste and lawn clippings (production time 18 h), observed that when producing aerated CTs, the Na, K and Mg content was higher in CTs produced from banana leaves (+195%, +170% and +118%, respectively), while the P and Cu content was higher in CTs produced from lawn clippings (+37% and +820%). Marın et al. [38], on the other hand, observed a 2.4-times higher ammonia content (mg/L) in un-aerated CTs produced from grape marc compost than in aerated CTs. Table 1 shows the chemical characterization of compost tea (CT) according to the different types of waste source and brewing process.

Table 1. Changes in the chemical composition of compost tea (CT) according to different brewing processes adopted.

Waste Material	CT Brewing Process				(dS m^{-1})		ppm													Reference
	Compost-to-Water Ratio	Production Time	Areated or Not	Temperature (°C)	pH	EC	NO^{3-}	P_2O_5	K_2O	SO_4^{2-}	Ca	Mg	Na	Cl	Humic Acids	Fe	Mn	Cu	Zn	B	
Banana leaves	1000 g compost:17 L water (tap water)	18 h	Aerated	-	6.62	4.19	92.5	80.7	47.03	-	443.7	150.1	28.55	-	-	-	-	0.05	0.39	-	[33]
		27 h	Aerated	-	6.5	4.47	116.2	89.3	38.24	-	439.5	240.1	19.05	-	-	-	-	0.06	0.52	-	
		36 h	Aerated	-	6.57	3.9	132.7	83.9	42.64	-	436.5	129.2	20.99	-	-	-	-	0.05	0.35	-	
		56 h	Not areated	-	6.12	4.46	368	104.4	40.44	-	151	105	21.53	-	-	-	-	0.03	0.08	-	
		112 h	Not areated	-	6.43	4.86	204.6	107.5	60.22	-	167	146.8	26.38	-	-	-	-	0.04	0.08	-	
		168 h	Not areated	-	6.18	4.58	352.9	104.6	45.93	-	250.8	174	21.38	-	-	-	-	0.04	0.08	-	
Cattle manure	1:10 (*w*:*v* with tap water)	48 h	Areated	10	9.1	10.5	-	-	-	-	-	-	-	-	-	-	-	1.12	1.66	-	[39]
				20	9.1	11.5	-	-	-	-	-	-	-	-	-	-	-	1.43	1.93	-	
				30	8.9	12.5	-	-	-	-	-	-	-	-	-	-	-	2.05	2.91	-	
		10 days	Not areated	10	9.1	11	-	-	-	-	-	-	-	-	-	-	-	1.68	2.46	-	
				20	8.8	12.5	-	-	-	-	-	-	-	-	-	-	-	2.08	3.14	-	
				30	8.4	14.7	-	-	-	-	-	-	-	-	-	-	-	2.17	2.01	-	
Chicken manure-based termophilic compost	1:10 (*v*:*v*)	12 h	Areated	-	7.3	22.2	289.2	-	-	-	152.6	138.3	-	-	94.9	-	-	-	-	-	[40]
Chicken manure-based vermicompost (aged)				-	6.8	3.4	137.9	-	-	-	59.6	61.6	-	-	464.8	-	-	-	-	-	
Chicken manure-based vermicompost (fresh)				-	6.9	1.4	39.6	-	-	-	38.7	33.3	-	-	435.3	-	-	-	-	-	

Table 1. *Cont.*

Waste Material	CT Brewing Process					(dS m^{-1})	ppm														Reference
	Compost-to-Water Ratio	Production Time	Areated or Not	Temperature (°C)	pH	EC	NO^{3-}	P_2O_5	K_2O	SO_4^{2-}	Ca	Mg	Na	Cl	Humic Acids	Fe	Mn	Cu	Zn	B	
Commercial compost (Almagro, Ciudad Real province, Spain, LOT 2010/02)	1:4 (*w*:*v*)	14 days	Aerated	25	7.88	5.44	193	14.2	2247	353	1758	128	292	174	-	64.59	2.69	0.52	2.32	-	[38]
Commercial compost (Almagro, Ciudad Real province, Spain, LOT 2010/02)			Not areated	25	8.24	5.34	76	14.9	1892	626	290	92	215	184	-	0.22	0.57	0.16	0.17	-	
Commercial compost (La Manchela, Albacete province, Spain, LOT 2010/04)			Aerated	25	7.79	9.36	168	11.2	2626	2723	967	272	585	177	-	5.91	0.98	0.12	0.58	-	
Commercial compost (La Manchela, Albacete province, Spain, LOT 2010/04)			Not areated	25	8.19	9.18	155	12.7	2467	2241	819	491	529	170	-	0.05	1.19	0.01	0.81	-	
Green and pruning waste consisting of a mixture of grass cuttings and pruning debris	1:5 (*v*:*v*)	5 days	Areated	20	7.32	1.22	3200	102	3840	28	79	150	-	-	190	-	-	-	-	-	[41]
		15 days		20	7.33	1.48	3300	368	4123	31	110	135	-	-	179	-	-	-	-	-	
		5 days		15	7.16	1.2	2741	61.4	2851.2	20	280	20	-	-	100.3	-	-	-	-	-	
		5 days		20	7.16	1.46	4700	105	4039	12	26	138	-	-	179	-	-	-	-	-	
		15 days		20	7.12	1.43	3934	315	4016	13	20	128	-	-	183	-	-	-	-	-	
		5 days		15	7.11	1.23	3050	56.2	2782.2	15	101	22	-	-	98.3	-	-	-	-	-	
Green waste termophilic compost	1:10 (*v*:*v*)	12 h	Areated	-	7.8	3.3	8.4	-	-	-	48.7	21.2	-	-	556.5	-	-	-	-	-	[40]
Lawn clippings	1000 g compost:17 L water (tap water)	18 h	Aerated	-	5.16	3.64	145.7	110.9	17.37		361.7	68.6	9.68	-	-	-	-	0.46	0.54	-	[29]
		27 h		-	5.13	3.73	140.1	106	11.87		516.4	75.6	9.7	-	-	-	-	0.046	0.62	-	
		36 h		-	5.07	4.04	126.1	109.1	17.37		553.2	74.2	7.47	-	-	-	-	0.046	0.76	-	
		56 h	Not areated	-	6.08	3.85	319.8	104.2	16.27		436.6	43.8	14.5	-	-	-	-	0.06	0.97	-	
		112 h		-	5.97	3.88	93.2	126.7	9.68		488.1	78.5	14.5	-	-	-	-	0.05	1.04	-	
		168 h		-	6.02	3.75	223	123.8	16.27		504.2	83	2.95	-	-	-	-	0.05	0.75	-	

Table 1. *Cont.*

Waste Material	CT Brewing Process					(dS m^{-1})	ppm														Reference
	Compost-to-Water Ratio	Production Time	Areated or Not	Temperature (°C)	pH	EC	NO^{3-}	P_2O_5	K_2O	SO_4^{2-}	Ca	Mg	Na	Cl	Humic Acids	Fe	Mn	Cu	Zn	B	
Mixture of horticultural crop residues, tree leaves and field crop waste with poultry litter and cow dung	1:10 (*p*:*v*) (Tap water)	7 days	Areated	-	7.93	2.93	-	-	-	-	-	-	-	-	-	-	-	-	-	-	[42]
Mixture of sheep manure, beef manure and sheep litter composed mainly of straw (year 2004)	1:10 (*w*:*v*)	72 h	Not areated	-	7.31	-	-	-	-	-	23.4	13	13.1	-	-	9.3	0.82	0.05	0.38	0.11	[43]
Mixture of sheep manure, beef manure and sheep litter composed mainly of straw (year 2005)				-	6.74	-	-	-	-	-	6.58	6.64	13.1	-	-	0.47	0.008	0.006	0.006	0.31	
Mixture of sheep manure, beef manure and sheep litter composed mainly of straw (year 2006)				-	6.68	-	-	-	-	-	41.5	35.7	29.2	-	-	0.82	0.11	0.06	0.08	0.38	
Stimol-C® (the organic feedstock consists of mixed plant materials, straw and composted cow manure from biological farms)	2:100 (*p*:*v* with tap water)	48 h	Aerated	-	6.4	1.37	1.4	18.4	-	170	-	-	-	-	-	2.3	-	-	-	-	[44]
Vitafert commercial compost	13:100	24 h	Aerated	-	6.96	3.38	199	-	-	-	437.4	39.5	-	-	-	0.05	-	0.09	0.03	2.35	[45]

The carbon-to-nitrogen (C/N) ratio also influences the amounts of fungi and bacteria in the waste materials. Waste materials with a low C/N ratio are characterized by a high amount of bacteria. In contrast, on the other hand, straw and dry leaves have a very high C/N ratio and therefore have a higher amount of fungi than bacteria. Comparing aerated and non-aerated CTs produced from composts based on spent mushroom substrate (A), grape marc (B), horticultural crops (C) and vermicompost (D), a higher microbial biodiversity was observed in the A-based aerated CTs and a higher number of microbial species was observed in the D-based aerated CTs. However, in C-based un-aerated CTs (although the lowest microbial richness and diversity was observed), the uniformity index was found to be the highest [27]. It is interesting to observe that CTs produced from coffee processing waste have high antioxidant activity due to both a high concentration of phenols in the coffee waste and the aromatic composition of the CTs obtained from this waste [25,46].

3.2. Compost Quality Indicators

The quality of a compost can be determined by how mature it is at the end of the composting process and the effects it has on plants. In general, it is possible to say that immature compost reduces germinability and crop growth. A good compost should not have a germinability index below 70%. Furthermore, if the starting compost has an excessive amount of heavy metals such as iron, lead, copper and cadmium, these can be released into the CT and absorbed by the crops, affecting both the environment and human health [14]. This section will describe in more detail the physical, chemical and biological parameters that help us and determine the state of maturity of a compost.

3.2.1. Physical Indicators

Physical indicators such as temperature, odor, color and moisture provide a preliminary qualitative assessment of compost maturity. However, these indicators are greatly influenced by external environmental variables, so they are currently used only as auxiliary indicators. A good quality compost must have undergone a complete maturation process characterized by a mesophilic phase, a thermophilic phase, a cooling phase, and a final maturation phase [47–49]. Each stage, therefore, as suggested by their name, will have to fall within certain temperature ranges to ensure the proliferation of some microorganisms over others and the complete transformation of organic matter into compost. However, with knowledge of temperature alone, it is not possible to determine the degree of maturity of the compost. In fact, during the composting process, a change in the color and smell of the compost can also be observed. Regarding color, very often it is possible to say that a compost is mature when it reaches a darker coloration than when it started. The dark coloration is determined by the process of the humification and mineralization of the organic matter [50,51]. However, the use of this parameter for evaluating the maturity of a compost is limited to light-colored starting materials only and is also subject to individual color perception. The feasibility in using CIELAB color variables as a quantitative indicator of color is currently being investigated [52]. In some cases, monitoring odor can also be useful in defining the degree of maturity of a compost. This is true when compost is produced from organic wastes that are rich in sulfur and nitrogen, where odor decreases as the compost matures. In the case where compost is produced from plant waste, there is no odor emission during the composting process. Although no unambiguous standard is established to quantify the moisture content of mature compost (due to the variability in the moisture content in the starting material and from the composting method), it is still possible to say that compost is maturing when its moisture content decreases over time.

3.2.2. Chemical Indicators

Although there are not necessarily marked differences between the pHs of a mature compost versus an immature one, it is worth noting that the ideal pH range of a good quality compost is between 5.5 and 8.5 [53,54]. Electrical conductivity (EC), on the other hand, should be less than or equal to 4 mS/cm [55,56].

The C/N ratio is a parameter that is closely related to the microbial activity in the compost. C/N ratios around 10–15 favor mineralization and influence the bioactive properties of the compost. C/N ratios around 35, on the other hand, immobilize nitrogen by transforming inorganic nitrogen into organic nitrogen [32]. Therefore, the C/N ratio (ratio of total carbon to total nitrogen in the solid phase), it is good if it is around 15 and 20 at the end of the composting process [57,58]. However, the C/N ratio cannot always correctly define the degree of maturity of a compost. This is the case for feedstocks with very high or very low C/N ratios where it would be better to use the water-soluble C/N ratio as an indicator. In fact, regardless of the raw material, compost is defined as mature when the water-soluble C/N ratio stabilizes around 5 or 6 [59]. Martínez-Yáñez et al. [32] state that during the production process of aerated CT, the C/N ratio decreases because of CO_2 production. Furthermore, these authors state that the bacterial microbiota changes according to the C/N ratio in the CT. Therefore, the C/N ratio could be used as a chemical indicator of bacterial metabolism in CT [32].

During the composting process, nitrifying microorganisms reduce the ammonium content and increase the nitrate content, so a NO_3^-/NH_4^+ ratio of less than 0.16 can be attributed to a mature compost [60,61]. The cation exchange capacity of a good quality compost should be above 60 meq.100 g^{-1} of organic matter [62,63].

It is also essential to assess the content of heavy metals and toxic compounds in the compost. This is because if their content is excessive, they could be released into the CT and become bioavailable for crops [9]. Currently in the European Union, the limits for heavy metal content in soil conditioners, expressed in mg/kg dry biomass, are for 1 for Cadmium (Cd), 100 for Chromium (Cr), 200 for Copper (Cu), 0.45 for Mercury (Hg), 40 for Nickel (Ni), 100 for Lead (Pb), 300 for Zinc (Zn) and 100 for Arsenic (As) (EU Decision 2022/1244). An excessive heavy metal content could negatively affect the photosynthetic activity of crops as well as the microbiological and enzymatic activity of the soil.

Since organic matter is humified during the composting process, the humification ratio (HA/FA) also determines the mature state of a compost, and its value should be greater than or equal to 1.9. This ratio tends to increase during the composting process as fulvic acid is only an intermediate step in the formation of humic acids [64]. Humic substances are a supramolecular aggregation of heterogeneous small molecules that can improve crop growth [65]. These are classified into humic acids, fulvic acids and humins according to their molecular weight [66]. Their composition is also closely related to the starting material [67]. Depending on the pH of the water used for CT production, humic substances can be extracted more or less efficiently.

In conclusion, it can be extrapolated from the above information that the starting material makes it difficult to standardize optimal ranges related to chemical parameters in defining the degree of maturity and quality of a compost. Therefore, it would be interesting to adopt differentiated standards based on the type of starting material.

3.2.3. Biological Indicators

Microbial respiration is one of the main biological indicators that helps determine the state of maturity of a compost. However, it is difficult to use this quality criteria. It is assumed that the microbial populations in the compost are beneficial to the soil and crop growth [27]. Additionally, microbial respiration decreases as the degree of decomposition

of organic matter increases [68,69]. Ryckeboer et al. [70] report that the Bacillus species is dominant during the thermophilic composting phase. The rate of oxygen uptake can be measured by some tests to monitor the state of maturity of organic matter [62,71]. Some authors claim that microbial diversity is the most important factor in determining the quality of CT. This diversity, however, depends on both the starting material used for the compost and the production processes of the CT [9]. In fact, it is shown through 16S metabarcoding analysis that the bacterial microbiota of CTs are significantly influenced by the material used for compost production [32]. From the results obtained by Diànez et al. [27], a dominance of *Bacteroidetes* and *Proteobacteria* can be observed in all the CTs analyzed. This, according to the authors, is probably due to the presence of these phylum already in the composts used for CT production. Martínez-Yáñez et al. [32] observed that the basic bacterial microbiota of the different aerated CTs included the phyla of *Firmicutes*, *Proteobacteria*, *Acidobacteria* and *Chloroflexi*. However, the C/N ratio of the CTs influenced both the abundance of phyla and the bacterial genera. Table 2 shows the main phyla and bacterial genera observed in aerated CTs by different authors.

Table 2. Bacteria most frequently found in compost teas.

Bacteria		
Phyla	**Genus**	**References**
Actinobacteria	*Micrococcus*	[9]
Bacillota	*Bacillus*	[9,28,32,72]
	Weissella	[32]
	Brevibacillus	[73]
	Staphylococcus	[9]
Proteobacteria	*Pseudomonas*	[9,32,72,73]
	Burkholderia	[9]
	Comamonas	[28,73]
Pseudomonadota	*Sphingobium*	[28,32,73]
	Azotobacter	[32]
Firmicutes	*Lactobacillus*	[9,28,32]
	Clostridium	[28,32,73]

3.3. Compost Tea: Compost/Water Ratio

The compost/water dilution ratio can vary from 1/1 to 1/50 [26]. Although there is no unambiguous guidance on the production of quality compost tea, many authors have obtained good results using a 1/10 (*v*/*v*) compost/water dilution ratio, which would ensure a continuous exchange between the two phases [9]. This compost/water ratio influences the concentration of nutrients and microorganisms in the CT. Comparing different dilution ratios of CT (fermented) produced from chicken manure (control, 1/5, 1/10, 1/20, 1/30 and 1/60), the best results in terms of the fresh weight of pak choi were obtained at the dilution ratio of 1/20 [74]. It is observed in fermented CT produced from agricultural waste that with increasing dilution ratios (1/5, 1/10, 1/20, 1/50 and 1/100), not only is the nitrogen (ppm), zinc (ppm) and iron (ppm) content reduced, but also the microbial concentration expressed in cfu/mL [75].

pH, Brewing Time, Temperature and Oxygenation

For the production of quality aerated CTs, several authors suggest a dissolved oxygen concentration of around 5.5 ppm (mg/L) and ambient temperature [9]. In the case of un-aerated compost teas, it is still uncertain whether or not a minimum level of oxygen is required. However, information regarding measurements of pH, temperature and dissolved oxygen very often goes unreported in scientific articles. The literature suggests that

increasing the temperature during the production of fermented CT could increase microbial activity. However, the desired range would be between 20 °C and 38 °C. Above the maximum temperature limit, the enzymes present in the compost could be denatured [76]. Indeed, a reduction in β-1,4-N-acetylglucosaminidase (NAG) enzyme activity has been observed in aerated and heated CT compared to the content present in aerated CT without heating [76]. Islam et al. [77] also add that at temperatures that are lower than optimal, the enzymes do not perform their action due to the solidification of the membranes. According to Ingham [19], the production time for aerated CTs ranges from 1 to 3 days while for non-aerated CTs it ranges from 3 to 5 days. However, there is a high level of variety in the literature on production times to produce CTs. There is no shortage of cases where the production process for un-aerated CTs takes up to 14 days [12]. In general, increasing the extraction time leads to a reduction in the content of organic carbon and total N in the CT, as well as bacterial and fungal populations [77]. In the study by Raza et al. [75], nitrogen and potassium concentrations (ppm) were compared in fermented CT produced from agricultural waste at 1-, 3-, 5-, 7-, 10-, 15- and 30-day intervals. Regarding nitrogen, its concentration increased to a maximum during the 5th day of fermentation and then decreased almost linearly until the 30th day of production. The potassium content, on the other hand, increases until day 10 of production, and then decreases linearly until day 30.

Lu et al. [73], on the other hand, analyzed the microbiota of a vermicompost tea after 24 and 48 h of production. These authors observed significant differences in the diversity of both bacterial and fungal communities in the CT at the two different time points. Increasing production time increases bacteria the number of belonging to *Clostridium* spp., *Sphingobium yanoikuyae*, *Pseudomonas* spp., *Brevibacillus choshinensis* and *Comamonas ferrigena*. In addition, increasing production hours also increases the number of fungal species *Morierella*, *Staphylotrichum* and *Chaetomium*. However, production processes longer than 24 h reduce the populations of several *Bacillus* species.

3.4. Compost Tea: Application Methods

Compost tea can be applied as is or diluted in different ratios with dechlorinated water. In case it is diluted, it is necessary to ensure that the number of microorganisms in the CT is not insufficient. Ingham [19] states that a good CT should contain approximately one billion bacteria per ml and 5–20 µg of total fungal biomass per ml. In the study by Brinton et al. [78], an increase in the number of aerobic bacteria is observed in CT obtained from a 3-day production process compared to CT obtained after 24 h of infusion. In this study, it is observed that an amount of aerobic bacteria in the CT between 10^8 and 10^{10} per ml improves the activity of the CT, promoting both greater solubilization of crop nutrients and greater resistance to fungal diseases.

Therefore, to ensure good foliar coverage or good soil wetting at the root system, it will be necessary to increase the number of applications or increase the number of liters given to the crops. Ingham [19], in their book, suggests to apply 1 L of CT (either as it is or diluted with water) to the soil when transplanting 10 cm high plants, making sure that the leaves and soil in the root system area are saturated. For foliar applications, it is suggested to apply 50 L/ha of CT when the plant height is 0 to 1.8 m.

Proper wetting is critical to ensure that beneficial microorganisms in the CT can hinder the growth of pathogens through both the occupation of all infection sites (on the leaf surface and root system) by beneficial microorganisms [12]. If the soil has a low indigenous microbial population, it is recommended to increase the volume or number of applications [19]. Ingham [19] suggests sampling soil to assess the microbial food web in order to prepare a suitable CT to return the missing microorganisms to the soil. After about three weeks, another soil sample should be taken to analyze the biomass and

microbial activity of the soil in order to decide if more CT applications should be made. In the study by Larkin [79], soil microbial populations and communities were evaluated with fatty acid methyl ester (FAME) profiles, dilution plates and substrate use profiles. For the determination of the soil microbiota, samples were taken 2, 10 and 24 weeks after transplanting the culture. Soils treated with aerated CT did not increase microbial populations in the first two weeks after application. However, microbial populations increased significantly with CT application compared to the control in later evaluations (after 10 and 24 weeks). A total of four CT treatments were carried out: before transplanting, during transplanting, 4 and 8 weeks after transplanting.

The application of CT in the root zone promotes both the release of nutrients into the soil and the decomposition of these nutrients by soil microorganisms, promoting crops' uptake. However, it should be considered that sufficient quantities of CT should be applied to reach the root zone in order to ensure the protection of the root system from pathogens [14]. Foliar application compared to root application, on the other hand, increases the opening time of stomata promoting the faster uptake of the nutrients contained in CT [80].

Furthermore, as already highlighted in the sub-section "pH, Brewing Time, Temperature and Oxygenation", different CT production times promote the growth of different microbial populations. Therefore, it would be profitable to apply CTs obtained from different production times in order to have a wide range of beneficial microorganisms for crop growth [73]. The last aspect to be considered is the variable effect of the CTs at different phenological crop stages. For example, in the study by Martínez-Yáñez et al. [32], the growth of tomato seedlings treated with three aerated CTs was significantly increased only by the CT with the C/N ratio of about 10.

Regarding the irrigation systems used for CT treatments, it is recommended to avoid sprinkler and drip irrigation systems. In order to avoid nozzle fouling, spreaders should be preferred for soil applications and surface irrigation systems should be preferred for foliar applications [77].

4. Opportunities and Threats of Compost Tea as a Biostimulant

In this section, the microorganisms and biomolecules with biostimulant action present in aqueous CT extracts will be investigated. A biostimulant according to the definition proposed by Du Jardin [81] is "Any substance or microorganism applied to plants with the aim of improving nutritional efficiency, tolerance to abiotic stresses and/or quality traits of crops regardless of nutrient content".

Biostimulants based on microorganisms that can interact both directly and indirectly with plants are called Plant Growth-Promoting Bacteria (PGPB). These microorganisms can synthesize growth regulators and can increase the bioavailability of some nutrients to crops, facilitating their uptake [82]. As shown in Table 2, the literature shows that the most common bacteria in CT belong to the phyla of *Actinobacteria*, *Bacillota*, *Proteobacteria*, *Pseudomonadota* and *Firmicutes*. Some of the genera belonging to these phyla, such as *Pseudomonas*, *Bacillus*, *Rhizobium* and *Azotobacter*, have been widely studied to understand their potential roles as biostimulants [82]. Specifically, species belonging to the *Bacillus* phyla promote crop root uptake through the production of metabolites capable of solubilizing soil nutrients [83]. In addition to this, some *Bacillus* species are able to supply N to crops by synthesizing atmospheric nitrogen [82]. It is also interesting to note that in cases where the concentration of N in the soil is excessive, the bacteria can utilize it for their own metabolism, mitigating the negative effects that could be seen on the growth of the crop's root system [82]. These bacteria have also been associated with the synthesis of hormones that can promote crop growth, such as cytokinins, auxins and gibberellins [83]. Within the phyla *Proteobacteria*, some species of the genus *Pseudomonas* are associated with

biostimulant activity, and are able to both improve nutrient uptake by plants and synthesize vitamins [82]. The *Azotobacteria* belonging to the phyla *Pseudomonadota*, on the other hand, are free-living bacteria that fix atmospheric nitrogen, making sources of inorganic N available to plants [32]. Bacteria belonging to the genus *Clostridium* (*Firmicutes*), in addition to fixing nitrogen as in the case of *Azotobacter*, can solubilize phosphates and reduce iron [73]. In general, there is a wide range of plant–microorganism interactions. These very often are bidirectional. In fact, under plant stress conditions (both those that are abiotic and biotic), it is the plant that releases root exudates as a signal to the microorganisms in the soil. The signal will induce the micro-organisms to reduce stress through the production of siderophores, N fixation or through the solubilization of P [84].

Secondary metabolites may also have a biostimulant action [85]. A number of molecules with potential biostimulant action have been identified in CTs, such as humic acids and phenolic compounds. Humic acids have long been known for their biostimulating action. These molecules are able to improve crop growth through both indirect and direct actions. It should be noted that humic acids are molecules consisting of carboxyl (R-COOH) and hydroxyl (R-OH) functional groups of aromatic rings. These functional groups appear to play an important role in crop nutrition by forming complex structures with nutrient cations [82]. Humic acids also interact with plant hormone signaling pathways and influence the expression of genes related to photosynthetic activity, phytohormones and cell metabolism [82]. The presence of humic acids in aqueous CT extracts, according to this review, was reported by [40,41,86].

Phenols are also associated with potential biostimulant activity due to their antioxidant activity. Their bioactivity is expressed through an increased scavenger capacity, which can inhibit the action and generation of free radicals [25]. In the study of Verrillo et al. [25], the antioxidant properties of various CT samples were analyzed, confirming that this activity is related to the total phenolic content of the CTs and more specifically to their aromatic composition.

However, CTs consist of a mixture of microorganisms and biomolecules. Therefore, the biological interactions between the different components, including possible antagonistic, additive or synergistic effects, need to be determined in more detail [85]. In Europe, the production and marketing of biostimulants are regulated by Regulation (EU) 2019/1009 (2019). In line with European regulations, the contaminant content in biostimulants must not exceed the threshold values of 1.5 mg/kg dry matter for Cd; 2 mg/kg dry matter for Cr; 120 mg/kg dry matter for Pb; 1 mg/kg dry matter for Hg; 50 mg/kg dry matter for Ni; 40 mg/kg dry matter for As; 600 mg/kg dry matter for Cu and 1500 mg/kg dry matter for Zn. Because a CT also consists of microorganisms, it should be considered that the pathogen content of a microbial biostimulant must not exceed the threshold levels of *Salmonella* spp., *Escherichia coli*, *Listeria monocytegens*, *Vibrio* spp., *Shigella* spp., *Staphylococcus auereus* and *Enterococcacecae* described in Regulation (EU) 2019/1009 (2019). In addition to safety requirements, the document Regulation (EU) 2019/1009 describes in detail the accepted methodologies for the production of a biostimulant. Regarding the registration of a biostimulant, information must be provided on the specific action of the biostimulant on the target crops, its method of storage, its application and expiry dates. Currently, CTs are excessively heterogeneous products and their biostimulant action on crops is not well defined. In the document Regulation (EU) 2019/1009, it should be noted that "The plant biostimulant must produce the effects declared on the label for the plants specified in it", which is currently difficult due to the impossibility of standardizing the final product. In addition, a CT is an aqueous solution rich in microorganisms, so if the final product is not sterilized, it would have to satisfy the European regulations for microbial plant biostimulants (CMC 7 of Part II of Annex II).

5. How Does Compost Tea Improve Crop Growth?

Numerous studies state that the exogenous application of compost tea can bring benefits ranging from the control of biotic agents to increasing the potential yield of various crops [29,87,88]. In this review, as previously reported, we are going to exclusively investigate the studies that have focused on evaluating the action of compost tea as a substance that can improve production performance in horticulture (Table 3).

Table 3. Effects of compost teas on growth performance of different horticultural species.

Crop	Growing Condition	Growing Media	Method of Brewing	Compost Material Used	Compost Tea Production Process Specifications	Application Method	Effect of Compost Tea	References
Bell pepper (*Capsicum annuum* L.)	Greenhouse	Loamy soil	Aerated water extraction	Waste of artichokes, wood chips, fennel and escarole	The duration of the fermentation process was one week, the compost tea obtained was refrigerated and stored at 4 °C. 1:5 compost–water dilution ratio	Foliar application	The application of compost tea singificantly improved the physiological status of treated plants and yield in two different years. Specifically, the treated plants were characterized by a higher number of fruits.	[89]
		Sand soil		Vegetable waste	The duration of the fermentation process was one week. Compost–water dilution ratio of 1:10.	Root application	In the two-year experiment, the application of compost tea detemined a significant increase in yield and quality parameters (vitamin C and carotenoid content).	[90]
		Peat-based substrate			The compost tea brew was extracted with a compost–water dilution ratio of 1:10.	Foliar application	Compared with control plants, the application of compost tea significantly increased vegetative growth, biometric parameters (length, diameter and fresh weight), total yield and plant nutritional status (N, P and K).	[91]
				Waste of tomato, escarole, wood chips, artichoke and fennel	The duration of the fermentation process was one week. Dilution ratio used was 1:5 compost–water.	Root and foliar application	Compared to control, compost tea application stimulated root growth of treated bell pepper seedlings by recording a significant increase in root length (+8%).	[92]
Kohlrabi (*Brassica oleracea*)	Greenhouse	Loam soil	Aerated water extraction	Waste of artichoke and fennel	The duration of the fermentation process was one week, the compost tea obtained was refrigerated stored at 4 °C. 1:5 Compost–water dilution ratio.	Foliar application	Compared to the control, the use of compost tea provided a significant increase in commercial yield (+32%) that is attributable to an improvement in the physiological status of treated plants.	[93]
Lettuce (*Lactuca sativa* L.)	Greenhouse	Coconut beat and Peat-based substrate	Aerated water extraction	Rice straw and Hinoki cypress bark	The duration of the fermentation process was 96 h at room temperature. Compost–water dilution ratio 1:10.	Root and foliar application	Regardless of the dose, application of aerated compost tea to the root zone increased shoot and root growth and lettuce yield.	[28]
		Loam soil		Waste of artichoke and fennel	The duration of the fermentation process was one week, the compost tea obtained was refrigerated and stored at 4 °C. 1:5 compost–water dilution ratio.	Root application	The use of compost tea positively influenced the physiological status of plants (significant increase in SPAD index) leading to an increase in commercial yield (+24%) compared to the control.	[93]
Melon (*Cucumis melo* L.)	Greenhouse	Peat-based substrate	Aerated water extraction	Waste of tomato, escarole, wood chips, artichoke and fennel	The duration of the fermentation process was one week. Dilution ratio used was 1:5 compost–water.	Root and foliar application	Compost tea application improved growth parameters (number of leaves and fresh biomass) of melon seedlings.	[92]

Table 3. *Cont.*

Crop	Growing Condition	Growing Media	Method of Brewing	Compost Material Used	Compost Tea Production Process Specifications	Application Method	Effect of Compost Tea	References
Pak choi (*Brassica rapa*)	Greenhouse	Peat-perlite	Aerated water extraction	Chicken manure, aged chicken manure, green waste and food waste	The compost tea brew was extracted with a compost–water dilution ratio of 1:10.	Root application	The different types of compost tea significantly differed in total nutrient, phytohormone and beneficial microorganism contents. The use of compost tea obtained from chicken manure-based vermicompost significantly influenced plant growth and development due to higher nitrogen and gibberellin contents.	[40]
Tomato (*Solanum lycopersicum* L.)	NT	Agar	Aerated water extraction	Green waste	The compost tea brew was extracted with a compost–water dilution ratio of 1:6.	Seed application	Application of compost tea significantly improved lateral root number and primary root length. Analyses performed on the compost tea used revealed relevant humic acid, nitrogen and amino acid contents.	[94]
	Greenhouse	Peat-based substrate		Pruning waste (branches and leaves of cypresses, willows and poplars).	The compost tea brew was extracted with a compost–water dilution ratio of 1:5. The fermentation period lasted 15 days at constant temperature (20 °C).	Root and foliar application	The application of compost tea increased the number of leaves, plant height and dry weight of plants compared with the control.	[41]
		Soil		Vegetable and animal wastes	The compost tea brew was extracted with a compost–water dilution ratio of 1:5.		Composted tea extract significantly improved the main parameters of vegetative growth (number of stem internodes, plant height and number of leaves), marketable yield and average fruit weight.	[95]
				Poultry manure, dried cow dung, green leaves, ash, top soil and dried crop residue	The duration of the fermentation process was 72 h. Compost–water dilution ratio 1:5.		Regardless of the mode of application, compost tea positively influenced the yield of tomato plants through a general improvement in nutritional status. However, the best results were recorded following root application.	[96]
		Peat-based substrate		Waste of tomato, escarole, wood chips, artichoke and fennel	The duration of the fermentation process was one week. Dilution ratio used was 1:5 compost–water.		Compared to control, compost tea application significantly increased root length (+9%) and stem diameter (+12%) of treated tomato seedlings.	[92]

The growing interest in more sustainable farming practices that are less dependent on "chemistry" makes the use of compost tea an attractive solution. The sought-after positive action of compost tea is mainly attributable to the co-presence of mineral elements (macro and microelements), valuable metabolites, humic acids and positive microorganisms [35,97]. This peculiar and heterogeneous formulation is responsible for the direct and indirect activation of biological and/or chemical mechanisms that make compost tea useful for improving agricultural production.

In the horticultural sector more than in many others, nursery activity is by definition the starting point [98]. Improving nursery quality would provide enormous benefits to the entire production chain. After assessing hygienic quality and phytotoxicity, Villecco et al. [92] evaluated in the nursery stage the application of seven different compost teas in three different horticultural species (tomato, bell pepper and melon). The results presented showed that the effect of compost tea on principal morphometric characteristics was strictly

type-dependent, highlighting that the starting matrix for compost tea production is a highly relevant and discriminate. To confirm this, Pant et al. [40] found, in a study on pak choi, how the agronomic performances of compost tea were closely dependent on the quality and composition of the starting compost. Specifically, the authors observed a better production response in plants treated with compost tea obtained from chicken manure-based vermicompost and chicken manure-based thermophilic compost compared to compost tea obtained from kitchen scraps and green waste. Through vermicomposting, high microbial abundance and diversity can be achieved, which would contribute to the release of substances that can inhibit the proliferation of microorganisms that are antagonistic to crops, while at the same time also promoting the triggering of crop defense mechanisms [11]. As suggested by the authors themselves, the improved bioactivity of the two compost teas obtained from chicken manure could be attributable to a higher presence of mineral nitrogen (N), live bacteria and giberellins that would stimulate root system growth contributing to the improved uptake of key nutrients.

The application of CTs in the root zone could also regulate soil pH, thus promoting greater enzyme activity related to biogeochemical nutrient cycling, which would contribute in making nutrients available for root uptake [99]. As mentioned in the previous sections, CTs show high protease and cellulase activity. These enzymes actively participate in the transformation of organic matter in the soil, converting nutrients into molecular forms that can be absorbed by crops. In one study, an increase in the alkaline phosphatase activity of CT-treated soil was also observed, which promoted the growth of cotton roots [100]. Similarly, the most interesting results obtained by Villecco et al. [92] were recorded after the foliar application of compost tea obtained from artichoke residues and combined fennel and artichoke residues. The better action of these compost teas, as claimed by the authors, could be attributable to a higher presence of N, a crucial element in the early stages of plant growth. As reported in the previous sections, one of the most abundant phyla found in various CTs is Proteobacteria. Bacteria in this phylum would contribute to increasing the nitrate concentration in the CT through the processes of ammonia oxidation into nitrate [27].

The exogenous application of compost teas in addition to increasing the dry and fresh biomass of seedlings positively influenced root growth. The latter aspect would be an interesting advantage, as seedlings with a more vigorous root system, in addition to benefiting from better anchorage in the soil, could respond better to the delicate transplanting phase, especially under suboptimal conditions such as water scarcity or nutrient limitation. As suggested by Lazcano and Domínguez [101] and Atiyeh et al. [102], the positive action exerted by compost tea cannot be attributed solely to the availability of key nutrients but also and especially to the presence of phytohormones and humic substances that characterize the composition of these products. Humic substances, for example, would increase the permeability of root cell membranes, which would improve both the uptake and transport of nitrates and the activation of genes involved in this activity [103].

With the aim of evaluating the action of compost tea on plant rhizogenesis, González-Hernández et al. [94] tested a compost tea obtained from green waste on agar-grown tomato seedlings. Although phytohormones were not detected, compared with the control, the treated tomato seedlings were characterized by a longer primary root length and more secondary roots. Excluding an improved nutritional supply (because the nutrition of control seedlings was standardized to the nitrogen content of compost tea) and the microbiological component (because the compost tea used was appropriately filtered to ensure aseptic conditions), the authors suggest that the application of compost tea may have simultaneously triggered direct and indirect signaling pathways related to the presence of secondary metabolites. Secondary metabolites would be produced by the microorganisms

in the CT and would therefore remain in the aqueous solution even after filtration. These would also be able to increase the ion exchange of nutrients in the circulating solution.

Aiming to investigate how a different application mode affects the action of compost tea, González-Hernández et al. [41] compared the vegetative growth of tomato plants following the foliar or root application of compost tea obtained from green residues. In contrast with what has been reported in the literature, the root application of compost tea improved vegetative and root growth while the foliar application did not result in significant differences compared to control plants. While the positive action of the root application could be related to the mineral endowment of the compost tea and the presence of humic substances that may have triggered biochemical responses such as the activation of ATPase in the plasma membrane [104], there being no effects recorded in the foliar-treated tomato plants could be attributable to the excessive concentration of the compost tea used, which was not diluted. The foliar application, in addition to being localized and undisturbed, takes advantage of stomatal openings and the presence of polar aqueous pores that allow for the faster adsorption of the elements in the compost tea so it is more effective than soil applications [80]. In light of this, González-Hernández et al. [41] hypothesized that the high concentration of compost tea used in their experiment caused the plants to reduce their stomatal openings, thus reducing the effectiveness of foliar application. This result underscores the relevance of compost tea dilution. Therefore, in order to avoid the under- or over-use of CT, more in-depth studies should be carried out on the choice of CT concentration to be used according to the application method considered. In addition to this, the frequency of application should also be investigated further.

Further comparison of the two different application methods was studied by Girshe et al. [96]. Both foliar and root applications at the higher doses (1200 mL per plant) improved the nutritional status and yield of tomato plants grown in the open field. The dose-dependent improvement in the production responses of the tested plants could be a consequence of the increased availability of nutrients and humic substances that are useful for sustaining plant growth. The literature shows that the organic molecules in the CT bind with the soil particles. In fact, due to the improved aggregation of soil particles, it is possible to achieve both greater water retention and a reduction in the leaching and evaporation of nutrients and water from the soil [14].

Girshe et al. [96] observed, following the application of compost tea, an advance in flowering that could be related to the hormonal action of compost tea. Indeed, the literature suggests that CT can contribute positively to flower formation by influencing plant metabolism and enzyme activity through the increased production of growth hormones [105]. In any case, even in this experiment, the highest yield performances were obtained from plants treated radically with compost tea followed by the foliar application and the control. These results were confirmed in a similar study conducted on greenhouse-grown tomatoes [95]. Specifically, Abubaker et al. [95] observed, compared with control plants, a general improvement in growth and better yield regardless of the mode of application of compost tea. Differently from root application, foliar application of compost tea significantly increased the number of fruits due probably to the better translocation of organic and non-organic compounds present in compost tea. The literature suggests that foliar applications of CT promote the adhesion of beneficial microorganisms and nutrients to the leaf surface, which would also promote greater tolerance to biotic stresses by reducing the space available for fungi and bacteria to adhere to the leaf surface [106]. Also in this experiment, the authors recorded significant early flowering. Despite the abundance of nutrients, the application of compost tea cannot be considered as an alternative strategy to synthetic fertilizers in integrated production settings but rather as a valuable tool to reduce their use. For the same N content, Li et al. [107] observed that compared with

inorganic fertilization, root application of compost tea significantly reduced vegetative biomass but did not reduce tomato yield. In any case, the most interesting results concerned the dynamics related to metabolism, transport and nitrogen accumulation in tomato fruits. Indeed, the authors found that the application of compost tea promotes the allocation of nitrogen (in NH_4^+ and amino acid form) in tomato fruits by incentivizing earlier senescence of green organs, which would justify the reduction in vegetative biomass recorded while encouraging greater biomass accumulation in the fruits.

The purpose of the study carried out by Kim et al. [28] was to investigate in detail the effects of different doses (0.1, 0.2, 0.4 and 0.8%) of compost tea on the vegetative growth of lettuce, maize and soybean grown in greenhouses. Regardless of the dose used, the application of compost tea significantly improved the growth and development of all three species. In any case, if for corn and soybean the best results were obtained using the highest dose (0.8%), for lettuce, the highest fresh weight was obtained from plants treated at the 0.4% dose. These results highlight how the choice of the optimal dose cannot be made without considering the interaction with the genotype. Xu et al. [108] report that CT (whether aerated or not) may be more phytotoxic to lettuce than the application of other compost extracts. Promotion effects mediated by the use of compost tea have also been documented on kohlrabi and lettuce production [93]. Although for both crops the application of compost tea significantly increased yield, the best results were recorded on kohlrabi, emphasizing once again the key role of genotype.

In a nutrient-limited context such as organic farming, the application of compost tea could be a valuable tool for farmers. Zaccardelli et al. [89] observed on organic bell pepper a significant increase in yield following the foliar application of compost tea. Although fruit size and weight were unchanged, a significant increase in fruit number was observed in the treated plants. In light of these results, Zaccardelli et al. [89] hypothesized that the positive action of compost tea stimulated a more balanced fruit set by reducing premature fruit drop. In any case, the treated plants benefited from a general improvement in nutritional and physiological status found in the highest leaf greenness index (Soil and Plant Analysis Development, SPAD). In agreement with this, Xu et al. [109] recorded on cucumber plants treated with compost tea an improvement in chlorophyll content. Both nutrients and microorganisms involved in the biogeochemical cycling of nutrients in the CT contribute to improving the nutritional status of crops. All this results in an increase in plant photosynthesis, which implicitly explains why the chlorophyll content of leaves increases [105]. Also on bell pepper, Mohamed et al. [90] observed, following the root application of compost tea obtained from vegetable source material, a significant improvement in yield and yield parameters (fruit number and weight) and nutritional quality (total soluble solids and carotenoids).

The results reported in the study by Abd-Alrahman et al. [91] on bell pepper, in addition to once again highlighting the positive role of compost tea, demonstrate the economic impact CT would have on the farmers' income. The improved yield provided by the application of compost tea justifies the significant improvement in farmers' net income by emphasizing not only environmental but also economic sustainability in the use of this product.

6. Conclusions and Future Perspectives

This review presents the most recent experimental evidence regarding the use of compost tea to optimize the growth of horticultural crops. From this paper, all authors adopted the aerated method for compost tea extraction. This is probably because compost tea fermentation is mainly adopted when the goal is to improve crop resistance to biotic stresses [26]. Most of the experiments reviewed were conducted in the greenhouse, and

both the effects attributed to foliar and root applications were observed. The positive effects induced by the foliar application of compost tea were related to the faster adsorption of nutrients in compost tea through stomatal openings [80]. Root applications of compost tea, on the other hand, can stimulate the proliferation of root absorptive [110]. However, it should be remembered that the effects explicated by compost tea can vary greatly depending on the material used for compost tea production, and by the interaction between application dose and the horticultural species to which it is being applied. Therefore, one of the main challenges consists of the difficulty of standardizing CTs due to their heterogeneity. Nevertheless, currently several manufacturing companies, especially in the United States, have formulated commercial compost tea products, indicating both their composition and their potential effects on crops [12]. However, it would be desirable to standardize the method of CT production and application not only nationally but also internationally, in order to achieve a standardized product quality that is able to guarantee specific performance [14]. However, the thing that would improve the environmental and economic sustainability of farms would be the self-production of compost tea. The self-production of CT at a farm level would improve the sustainability of the horticultural supply chain by reducing not only the amount of waste produced but also the use of production inputs such as pesticides and synthetic fertilizers. CT-based formulations could also contribute to improving the economic and environmental sustainability of organic farms because of their microbiological and nutritional component that promotes crop growth [9]. For this reason, self-production should be a more-encouraged practice because it could enhance the value of farm waste in a circular economic perspective. Not only would this approach reduce the costs associated with the disposing of waste in landfills, but it would also allow the identification of specific cultivation protocols that take into consideration both the nature of the farm waste used and the species to which compost tea would be applied. Especially when the possibility of self-producing CT at a farm level is considered, it is recommended that farmers attend training courses. This will also help to standardize CT farm self-production methods. Several CT studies have been carried out precisely with the aim of providing more useful information in defining quality standards. Verrillo et al. [25] determined the molecular characteristics of three different compost teas isolated from farm composts, with the aim of evaluating the relationship between the structure and bioactivity of molecules dissolved in the compost teas. Their results show that the antioxidant and antimicrobial activity of the compost teas was related to the amount and structural complexity of the bioavailable aromatic compounds, while the hydrophobic composition of the compost teas influences their bioactive properties through the formation of pseudo-micelles that are capable of transporting bioactive molecules close to the root membranes. Despite the growing interest in characterizing CT in order to define quality standards, it is not always monitored. The factors to be considered for a comprehensive assessment of the quality of a CT must take into account the content of heavy metals, organic contaminants [9], potential contamination by human and plant pathogens [12], and a characterization of the microbiome and active compounds contained in the CT. The characterization aims to help identify key substances and microorganisms that are responsible for specific promoters of crop growth and soil health. However, the understanding of the microbial populations in the compost and its aqueous extracts is currently rather limited. Indeed, polymerase chain reaction (PCR)-based analyses of 16S and 18S rRNA amplicons do not provide sufficient information on the compost and CT microbiome due to its semi-quantitative character [33]. The multi-omic approach should be used for this type of analysis. Genomic analyses allow the entire genomes of the microorganisms in the sample to be analyzed, providing an overview of the complexity and heterogeneity of microorganisms and the genes linked to specific functions. However, with genomic analyses, it is not possible to know whether

these genes are active or not. It is, in fact, transcriptomic analyses that make it possible to know which of the genes are active. However, these analyses do not provide information on the transcripts being translated into proteins. With proteomic analyses, on the other hand, it is possible to characterize the protein composition of the sample's microbiota in order to understand which metabolic pathways are active. However, this analysis is not efficient when there are complex microbial communities. Thanks to metabolomics, however, it is possible to reveal the metabolites released by microorganisms, providing information on the interactions between the environment and microorganisms [33]. Through a multi-omics approach and by enhancing the knowledge related to these analyses, it will be possible to develop more performing CTs.

Therefore, a closer collaboration between private farms and the research sector, as previously said, is recommended to develop specific protocols that consider the molecular characterization of the compost tea that is obtainable on a farm and the target crop. This could maximize the benefits to crop growth and reduce the use of synthetic fertilizers, promoting more sustainable farming practices and optimizing on-farm resources.

Author Contributions: E.C., writing—original draft preparation; E.C., M.C., M.L., Y.R. and S.D.P., writing—review and editing. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: Not applicable.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Curadelli, F.; Alberto, M.; Uliarte, E.M.; Combina, M.; Funes-Pinter, I. Meta-analysis of yields of crops fertilized with compost tea and anaerobic digestate. *Sustainability* **2023**, *15*, 1357. [CrossRef]
2. Galindo, F.S.; Pagliari, P.H.; da Silva, E.C.; de Lima, B.H.; Fernandes, G.C.; Thiengo, C.C.; Bernardes, J.V.S.; Jalal, A.; da Silva Oliveira, C.E.; dos Santos, G.D. Unveiling contribution and fate of nitrogen with 15N techniques affected by microbial co-inoculation on field-grown maize: A novel approach to optimize N-fertilizer use efficiency. *Plant Physiol. Biochem.* **2024**, *217*, 109261. [CrossRef]
3. Adzawla, W.; Setsoafia, E.D.; Setsoafia, E.D.; Amoabeng-Nimako, S.; Atakora, W.K.; Camara, O.; Jemo, M.; Bindraban, P.S. Fertilizer use efficiency and economic viability in maize production in the Savannah and transitional zones of Ghana. *Front. Sustain. Food Syst.* **2024**, *8*, 1340927. [CrossRef]
4. Coskun, D.; Britto, D.T.; Shi, W.; Kronzucker, H.J. Nitrogen transformations in modern agriculture and the role of biological nitrification inhibition. *Nat. Plants* **2017**, *3*, 17074. [CrossRef] [PubMed]
5. Montanarella, L.; Panagos, P. The relevance of sustainable soil management within the European Green Deal. *Land Use Policy* **2021**, *100*, 104950. [CrossRef]
6. Khan, M.N.; Sial, T.A.; Ali, A.; Wahid, F. Impact of Agricultural Wastes on Environment and Possible Management Strategies. In *Frontier Studies in Soil Science*; Springer: Berlin/Heidelberg, Germany, 2024; pp. 79–108.
7. Laufenberg, G.; Kunz, B.; Nystroem, M. Transformation of vegetable waste into value added products: (A) the upgrading concept; (B) practical implementations. *Bioresour. Technol.* **2003**, *87*, 167–198. [CrossRef]
8. Eldridge, S.M.; Yin Chan, K.; Donovan, N.J.; Saleh, F.; Orr, L.; Barchia, I. Agronomic and economic benefits of green-waste compost for peri-urban vegetable production: Implications for food security. *Nutr. Cycl. Agroecosyst.* **2018**, *111*, 155–173. [CrossRef]
9. De Corato, U. Agricultural waste recycling in horticultural intensive farming systems by on-farm composting and compost-based tea application improves soil quality and plant health: A review under the perspective of a circular economy. *Sci. Total Environ.* **2020**, *738*, 139840. [CrossRef]
10. Khair Biek, S.; Khudur, L.S.; Ball, A.S. Challenges and Remediation Strategies for Per-and Polyfluoroalkyl Substances (PFAS) Contamination in Composting. *Sustainability* **2024**, *16*, 4745. [CrossRef]
11. Baum, C.; Eichler-Löbermann, B.; Hrynkiewicz, K. Impact of organic amendments on the suppression of Fusarium wilt. In *Organic Amendments and Soil Suppressiveness in Plant Disease Management*; Springer: Berlin/Heidelberg, Germany, 2015; pp. 353–362.
12. Garg, J.; Rakshit, A. Compost Tea: An Emerging Nature-Based Supplement Strengthening Options for Durable Agriculture. *J. Soil Sci. Plant Nutr.* **2024**, *24*, 8075–8098. [CrossRef]

13. Bouchtaoui, E.M.; Haouas, A.; Dababat, A.A.; Lahlali, R.; Benali, A.; Fahr, M.; Smouni, A.; Azim, K.; Liu, Z.; Li, J. Exploring mechanisms of compost-mediated suppression of plant pathogens: A critical review. *Appl. Soil Ecol.* **2024**, *203*, 105644. [CrossRef]
14. Yin, J.; Wang, J.; Zhao, L.; Cui, Z.; Yao, S.; Li, G.; Yuan, J. Compost tea: Preparation, utilization mechanisms, and agricultural applications potential—A comprehensive review. *Environ. Technol. Innov.* **2025**, *38*, 104137. [CrossRef]
15. Szekely, I.; Jijakli, M.H. Bioponics as a promising approach to sustainable agriculture: A review of the main methods for producing organic nutrient solution for hydroponics. *Water* **2022**, *14*, 3975. [CrossRef]
16. Proietti, P.; Calisti, R.; Gigliotti, G.; Nasini, L.; Regni, L.; Marchini, A. Composting optimization: Integrating cost analysis with the physical-chemical properties of materials to be composted. *J. Clean. Prod.* **2016**, *137*, 1086–1099. [CrossRef]
17. Matouk, A.; EL-Hadidi, Y.; Tharwat, A.; Khafagy, S. Production of Compost Tea from Farm Wastes. *J. Soil Sci. Agric. Eng.* **2017**, *8*, 323–329. [CrossRef]
18. Pergola, M.; Maffia, A.; Carlucci, G.; Persiani, A.; Palese, A.M.; Zaccardelli, M.; Altieri, G.; Celano, G. An environmental and economic analysis of strawberry production in southern Italy. *Agriculture* **2023**, *13*, 1705. [CrossRef]
19. Ingham, E.R. *The Compost Tea Brewing Manual*; Unisun Communications: Corvallis, OR, USA, 2000.
20. Palmer, A.; Evans, K.; Metcalf, D. Characters of aerated compost tea from immature compost that limit colonization of bean leaflets by Botrytis cinerea. *J. Appl. Microbiol.* **2010**, *109*, 1619–1631. [CrossRef]
21. Palese, A.M.; Pane, C.; Villecco, D.; Zaccardelli, M.; Altieri, G.; Celano, G. Effects of organic additives on chemical, microbiological and plant pathogen suppressive properties of aerated municipal waste compost teas. *Appl. Sci.* **2021**, *11*, 7402. [CrossRef]
22. D'Amato, S.; Serio, A.; López, C.C.; Paparella, A. Hydrosols: Biological activity and potential as antimicrobials for food applications. *Food Control* **2018**, *86*, 126–137. [CrossRef]
23. St. Martin, C. Potential of compost tea for suppressing plant diseases. *CABI Rev.* **2015**, 1–38. [CrossRef]
24. Guaadaoui, A.; Benaicha, S.; Elmajdoub, N.; Bellaoui, M.; Hamal, A. What is a bioactive compound? A combined definition for a preliminary consensus. *Int. J. Nutr. Food Sci.* **2014**, *3*, 174–179. [CrossRef]
25. Verrillo, M.; Salzano, M.; Cozzolino, V.; Spaccini, R.; Piccolo, A. Bioactivity and antimicrobial properties of chemically characterized compost teas from different green composts. *Waste Manag.* **2021**, *120*, 98–107. [CrossRef] [PubMed]
26. Scheuerell, S.; Mahaffee, W. Compost tea: Principles and prospects for plant disease control. *Compost. Sci. Util.* **2002**, *10*, 313–338. [CrossRef]
27. Diánez, F.; Marín, F.; Santos, M.; Gea, F.J.; Navarro, M.J.; Piñeiro, M.; González, J.M. Genetic analysis and in vitro enzymatic determination of bacterial community in compost teas from different sources. *Compost. Sci. Util.* **2018**, *26*, 256–270. [CrossRef]
28. Kim, M.J.; Shim, C.K.; Kim, Y.K.; Hong, S.J.; Park, J.H.; Han, E.J.; Kim, J.H.; Kim, S.C. Effect of aerated compost tea on the growth promotion of lettuce, soybean, and sweet corn in organic cultivation. *Plant Pathol. J.* **2015**, *31*, 259. [CrossRef]
29. St. Martin, C.; Dorinvil, W.; Brathwaite, R.; Ramsubhag, A. Effects and relationships of compost type, aeration and brewing time on compost tea properties, efficacy against Pythium ultimum, phytotoxicity and potential as a nutrient amendment for seedling production. *Biol. Agric. Hortic.* **2012**, *28*, 185–205.
30. Tóthné Bogdányi, F.; Boziné Pullai, K.; Doshi, P.; Erdős, E.; Gilián, L.D.; Lajos, K.; Leonetti, P.; Nagy, P.I.; Pantaleo, V.; Petrikovszki, R. Composted municipal green waste infused with biocontrol agents to control plant parasitic nematodes—A review. *Microorganisms* **2021**, *9*, 2130. [CrossRef]
31. Sharma, A.; Soni, R.; Soni, S.K. From waste to wealth: Exploring modern composting innovations and compost valorization. *J. Mater. Cycles Waste Manag.* **2024**, *26*, 20–48. [CrossRef]
32. Martínez-Yáñez, M.G.; Silva-Ortega, C.O.; Hernández-Aranda, V.A.; Vallejo-Pérez, M.R.; Alcalá-Briseño, R.; Vega-Manriquez, D.X.; Aguilar-Benítez, G.; Jarquin-Gálvez, R.; Lara-Ávila, J.P. Analysis of bacterial microbiota of aerated compost teas and effect on tomato growth. *Microb. Ecol.* **2023**, *86*, 959–972. [CrossRef]
33. St. Martin, C.C.; Rouse-Miller, J.; Barry, G.T.; Vilpigue, P. Compost and compost tea microbiology: The "-omics" era. In *Biology of Composts*; Springer: Berlin/Heidelberg, Germany, 2020; pp. 3–30.
34. Seddigh, S.; Kiani, L. Evaluation of different types of compost tea to control rose powdery mildew (*Sphaerotheca pannosa* var. rosae). *Int. J. Pest Manag.* **2018**, *64*, 178–184. [CrossRef]
35. Pilla, N.; Tranchida-Lombardo, V.; Gabrielli, P.; Aguzzi, A.; Caputo, M.; Lucarini, M.; Durazzo, A.; Zaccardelli, M. Effect of compost tea in horticulture. *Horticulturae* **2023**, *9*, 984. [CrossRef]
36. Shaban, H.; Fazeli-Nasab, B.; Alahyari, H.; Alizadeh, G.; Shahpesandi, S. An Overview of the Benefits of Compost tea on Plant and Soil Structure. *Adv. Biores.* **2015**, *6*, 154–158.
37. Ngakou, A.; Koehles, H.; Ngueliaha, H. The role of cow dung and kitchen manure composts and their non-aerated compost teas in reducing the incidence of foliar diseases of *Lycopersicon esculentum* (Mill). *Int. J. Agric. Res. Innov. Technol. (IJARIT)* **2014**, *4*, 88–97. [CrossRef]
38. Marín, F.; Santos, M.; Diánez, F.; Carretero, F.; Gea, F.J.; Yau, J.A.; Navarro, M.J. Characters of compost teas from different sources and their suppressive effect on fungal phytopathogens. *World J. Microbiol. Biotechnol.* **2013**, *29*, 1371–1382. [CrossRef]

39. Carballo, T.; Gil, M.; Calvo, L.; Morán, A. The influence of aeration system, temperature and compost origin on the phytotoxicity of compost tea. *Compost. Sci. Util.* **2009**, *17*, 127–139. [CrossRef]
40. Pant, A.P.; Radovich, T.J.; Hue, N.V.; Paull, R.E. Biochemical properties of compost tea associated with compost quality and effects on pak choi growth. *Sci. Hortic.* **2012**, *148*, 138–146. [CrossRef]
41. González-Hernández, A.I.; Gómez-Sánchez, M.Á.; Pérez-Sánchez, R.; Morales-Corts, M.R. Garden Waste Compost Tea: A Horticultural Alternative to Promote Plant Growth and Root Traits in Tomato (*Solanum lycopersicum* L.) Plants. *Horticulturae* **2023**, *9*, 1127. [CrossRef]
42. Abdel-Haleem, E.-S.; Farrag, H.M.; Abeer, B.; Abdelrasheed, K.G. Combined use of compost, compost tea, and vermicompost tea improves soil properties, and growth, yield, and quality of (*Allium cepa* L.). *Not. Bot. Horti Agrobot. Cluj-Napoca* **2022**, *50*, 12565.
43. Hargreaves, J.C.; Adl, M.S.; Warman, P.R. Are compost teas an effective nutrient amendment in the cultivation of strawberries? Soil and plant tissue effects. *J. Sci. Food Agric.* **2009**, *89*, 390–397. [CrossRef]
44. Bonanomi, G.; Idbella, A.; Amoroso, G.; Iacomino, G.; Gherardelli, M.; De Sio, A.; Saccocci, F.; Abd-ElGawad, A.M.; Moreno, M.; Idbella, M. Agronomic impacts of chemically and microbiologically characterized compost tea in Mediterranean volcanic soils. *Front. Plant Sci.* **2025**, *16*, 1524884. [CrossRef]
45. Hirzel, J.; Cerda, F.; Millas, P.; France, A. Compost tea effects on production and extraction of nitrogen in ryegrass cultivated on soil amended with commercial compost. *Compost. Sci. Util.* **2012**, *20*, 97–104. [CrossRef]
46. Vuolo, M.M.; Lima, V.S.; Junior, M.R.M. Phenolic compounds: Structure, classification, and antioxidant power. In *Bioactive Compounds*; Elsevier: Amsterdam, The Netherlands, 2019; pp. 33–50.
47. Jain, M.S.; Jambhulkar, R.; Kalamdhad, A.S. Biochar amendment for batch composting of nitrogen rich organic waste: Effect on degradation kinetics, composting physics and nutritional properties. *Bioresour. Technol.* **2018**, *253*, 204–213. [CrossRef] [PubMed]
48. Abdellah, Y.A.Y.; Shi, Z.-J.; Sun, S.-S.; Luo, Y.-S.; Yang, X.; Hou, W.-T.; Wang, R.-L. An assessment of composting conditions, humic matters formation and product maturity in response to different additives: A meta-analysis. *J. Clean. Prod.* **2022**, *366*, 132953. [CrossRef]
49. Finore, I.; Feola, A.; Russo, L.; Cattaneo, A.; Di Donato, P.; Nicolaus, B.; Poli, A.; Romano, I. Thermophilic bacteria and their thermozymes in composting processes: A review. *Chem. Biol. Technol. Agric.* **2023**, *10*, 7. [CrossRef]
50. Palaniveloo, K.; Amran, M.A.; Norhashim, N.A.; Mohamad-Fauzi, N.; Peng-Hui, F.; Hui-Wen, L.; Kai-Lin, Y.; Jiale, L.; Chian-Yee, M.G.; Jing-Yi, L. Food waste composting and microbial community structure profiling. *Processes* **2020**, *8*, 723. [CrossRef]
51. Shan, Y.N.; Chen, J.H.; Wang, L.; Li, F.; Fu, X.H.; Le, Y.Q. Influences of adding easily degradable organic waste on the minimization and humification of organic matter during straw composting. *J. Environ. Sci. Health Part B* **2013**, *48*, 384–392. [CrossRef] [PubMed]
52. Rashwan, M.; Alkoaik, F.; Abdel-Razzak, H.; Ibrahim, M.; Fulleros, R.; Shady, M.; Abdel-Ghany, A. Evaluation of tomato waste compost stability and maturity using CIELAB color indicator. *J. Plant Nutr.* **2020**, *43*, 1427–1437. [CrossRef]
53. Sánchez-Monedero, M.; Roig, A.; Paredes, C.; Bernal, M. Nitrogen transformation during organic waste composting by the Rutgers system and its effects on pH, EC and maturity of the composting mixtures. *Bioresour. Technol.* **2001**, *78*, 301–308. [CrossRef]
54. Vázquez, M.; Sen, R.; Soto, M. Physico-chemical and biological characteristics of compost from decentralised composting programmes. *Bioresour. Technol.* **2015**, *198*, 520–532. [CrossRef]
55. Chung, W.J.; Chang, S.W.; Chaudhary, D.K.; Shin, J.; Kim, H.; Karmegam, N.; Govarthanan, M.; Chandrasekaran, M.; Ravindran, B. Effect of biochar amendment on compost quality, gaseous emissions and pathogen reduction during in-vessel composting of chicken manure. *Chemosphere* **2021**, *283*, 131129. [CrossRef]
56. Mussa, S.; Farhan, M.; Ahmad, S.; Zahra, K.; Kanwal, A.; Khan, Q.F.; Afzaal, M.; Wahid, A.; Sarker, P.K.; El-Sheikh, M.A. Exploring the utility of different bulking agents for speeding up the composting process of household kitchen waste. *Sci. Rep.* **2025**, *15*, 2488. [CrossRef] [PubMed]
57. Cheng, K.H.; Huang, M.C.; Lu, M.F.; Chou, Y.J.; Lin, J.J.M. Assessment of degree of maturity of compost produced by different kitchen waste composting methods. *Adv. Mater. Res.* **2013**, *652*, 1642–1651. [CrossRef]
58. Guo, R.; Li, G.; Jiang, T.; Schuchardt, F.; Chen, T.; Zhao, Y.; Shen, Y. Effect of aeration rate, C/N ratio and moisture content on the stability and maturity of compost. *Bioresour. Technol.* **2012**, *112*, 171–178. [CrossRef] [PubMed]
59. Chanyasak, V.; Kubota, H. Carbon/organic nitrogen ratio in water extract as measure of composting degradation. *J. Ferment. Technol.* **1981**, *59*, 215–219.
60. Bernal, M.; Sanchez-Monedero, M.; Paredes, C.; Roig, A. Carbon mineralization from organic wastes at different composting stages during their incubation with soil. *Agric. Ecosyst. Environ.* **1998**, *69*, 175–189. [CrossRef]
61. Noor, R.S.; Shah, A.N.; Tahir, M.B.; Umair, M.; Nawaz, M.; Ali, A.; Ercisli, S.; Abdelsalam, N.R.; Ali, H.M.; Yang, S.H. Recent Trends and Advances in Additive-Mediated Composting Technology for Agricultural Waste Resources: A Comprehensive Review. *ACS Omega* **2024**, *9*, 8632–8653. [CrossRef] [PubMed]
62. Azim, K.; Soudi, B.; Boukhari, S.; Perissol, C.; Roussos, S.; Thami Alami, I. Composting parameters and compost quality: A literature review. *Org. Agric.* **2018**, *8*, 141–158. [CrossRef]

63. Aparna, C.; Saritha, P.; Himabindu, V.; Anjaneyulu, Y. Techniques for the evaluation of maturity for composts of industrially contaminated lake sediments. *Waste Manag.* **2008**, *28*, 1773–1784. [CrossRef]
64. Jimenez, E.I.; García, V.P. Determination of maturity indices for city refuse composts. *Agric. Ecosyst. Environ.* **1992**, *38*, 331–343. [CrossRef]
65. Piccolo, A. The supramolecular structure of humic substances: A novel understanding of humus chemistry and implications in soil science. *Adv. Agron.* **2002**, *75*, 57–134.
66. Gautam, R.K.; Navaratna, D. Humic Substances: Its Toxicology, Chemistry and Biology Associated. In *Humic Substances*; InTech Open: Rijeka, Croatia, 2021; p. 97.
67. Mahler, C.F.; Dal Santo Svierzoski, N.; Bernardino, C.A.R. *Chemical Characteristics of Humic Substances in Nature*; InTech Open: Rijeka, Croatia, 2021.
68. Mengqi, Z.; Shi, A.; Ajmal, M.; Ye, L.; Awais, M. Comprehensive review on agricultural waste utilization and high-temperature fermentation and composting. *Biomass Convers. Biorefin.* **2021**, *13*, 5445–5468. [CrossRef]
69. Kong, Y.; Zhang, J.; Zhang, X.; Gao, X.; Yin, J.; Wang, G.; Li, J.; Li, G.; Cui, Z.; Yuan, J. Applicability and limitation of compost maturity evaluation indicators: A review. *Chem. Eng. J.* **2024**, *489*, 151386. [CrossRef]
70. Ryckeboer, J.; Mergaert, J.; Coosemans, J.; Deprins, K.; Swings, J. Microbiological aspects of biowaste during composting in a monitored compost bin. *J. Appl. Microbiol.* **2003**, *94*, 127–137. [CrossRef]
71. Gea, T.; Barrena, R.; Artola, A.; Sánchez, A. Monitoring the biological activity of the composting process: Oxygen uptake rate (OUR), respirometric index (RI), and respiratory quotient (RQ). *Biotechnol. Bioeng.* **2004**, *88*, 520–527. [CrossRef] [PubMed]
72. Pane, C.; Palese, A.M.; Spaccini, R.; Piccolo, A.; Celano, G.; Zaccardelli, M. Enhancing sustainability of a processing tomato cultivation system by using bioactive compost teas. *Sci. Hortic.* **2016**, *202*, 117–124. [CrossRef]
73. Lu, S.; Pentico, D.; Castro, R.; Dinh, S.; Love, J.J.; Larom, D.L.; Pérez, R.L.; Liu, C. Effect of ultraviolet light exposure and compost tea supplementation on growth, antioxidant activities, and microbiome of hydroponically grown mustard greens. *ACS Agric. Sci. Technol.* **2022**, *2*, 521–533. [CrossRef]
74. Wang, X.; Wei, Q.; Zhao, Q.; Liu, X.; Deng, H.; Li, Z. Short-chain fatty acid producers in compost tea as affected by brewing time and aeration condition. *J. Soils Sediments* **2023**, *23*, 3096–3107. [CrossRef]
75. Raza, A.; Ali, Z.; Ali, S.A. Standardization of protocol for the formulation of compost tea and its efficacy study on potato. *Pure Appl. Biol. (PAB)* **2023**, *12*, 1044–1055. [CrossRef]
76. Vail, D.C.; Hernández, D.L.; Velis, E.; Wills, A. Compost tea production methods affect soil nitrogen and microbial activity in a northern highbush blueberry system. *Agroecol. Sustain. Food Syst.* **2020**, *44*, 1370–1383. [CrossRef]
77. Islam, M.; Yaseen, T.; Traversa, A.; Kheder, M.B.; Brunetti, G.; Cocozza, C. Effects of the main extraction parameters on chemical and microbial characteristics of compost tea. *Waste Manag.* **2016**, *52*, 62–68. [CrossRef]
78. Brinton, W.; Storms, P.; Evans, E.; Hill, J. Compost teas: Microbial hygiene and quality in relation to method of preparation. *Biodynamics* **2004**, *2*, 36–45.
79. Larkin, R.P. Relative effects of biological amendments and crop rotations on soil microbial communities and soilborne diseases of potato. *Soil Biol. Biochem.* **2008**, *40*, 1341–1351. [CrossRef]
80. Mahmoud, E.; El-Gizawy, E.; Geries, L. Effect of compost extract, N2-fixing bacteria and nitrogen levels applications on soil properties and onion crop. *Arch. Agron. Soil Sci.* **2015**, *61*, 185–201. [CrossRef]
81. Du Jardin, P. Plant biostimulants: Definition, concept, main categories and regulation. *Sci. Hortic.* **2015**, *196*, 3–14. [CrossRef]
82. Baltazar, M.; Correia, S.; Guinan, K.J.; Sujeeth, N.; Bragança, R.; Gonçalves, B. Recent advances in the molecular effects of biostimulants in plants: An overview. *Biomolecules* **2021**, *11*, 1096. [CrossRef]
83. Radhakrishnan, R.; Hashem, A.; Abd_Allah, E.F. Bacillus: A biological tool for crop improvement through bio-molecular changes in adverse environments. *Front. Physiol.* **2017**, *8*, 667. [CrossRef]
84. Lyu, D.; Backer, R.; Subramanian, S.; Smith, D.L. Phytomicrobiome coordination signals hold potential for climate change-resilient agriculture. *Front. Plant Sci.* **2020**, *11*, 634. [CrossRef]
85. Li, J.; Lardon, R.; Mangelinckx, S.; Geelen, D. A practical guide to the discovery of biomolecules with biostimulant activity. *J. Exp. Bot.* **2024**, *75*, 3797–3817. [CrossRef]
86. Morales-Corts, M.R.; Pérez-Sánchez, R.; Gómez-Sánchez, M.Á. Efficiency of garden waste compost teas on tomato growth and its suppressiveness against soilborne pathogens. *Sci. Agric.* **2018**, *75*, 400–409. [CrossRef]
87. Mengesha, W.; Gill, W.; Powell, S.; Evans, K.; Barry, K. A study of selected factors affecting efficacy of compost tea against several fungal pathogens of potato. *J. Appl. Microbiol.* **2017**, *123*, 732–747. [CrossRef]
88. Diánez, F.; Santos, M.; Boix, A.; De Cara, M.; Trillas, I.; Avilés, M.; Tello, J. Grape marc compost tea suppressiveness to plant pathogenic fungi: Role of siderophores. *Compost. Sci. Util.* **2006**, *14*, 48–53. [CrossRef]
89. Zaccardelli, M.; Pane, C.; Villecco, D.; Palese, A.M.; Celano, G. Compost tea spraying increases yield performance of pepper (*Capsicum annuum* L.) grown in greenhouse under organic farming system. *Ital. J. Agron.* **2018**, *13*, 229–234. [CrossRef]

90. Mohamed, M.H.; Sami, R.; Al-Mushhin, A.A.; Ali, M.M.E.; El-Desouky, H.S.; Ismail, K.A.; Khalil, R.; Zewail, R.M. Impacts of effective microorganisms, compost tea, fulvic acid, yeast extract, and foliar spray with seaweed extract on sweet pepper plants under greenhouse conditions. *Plants* **2021**, *10*, 1927. [CrossRef]
91. Abd-Alrahman, H.A.; Aboud, F.S. Response of sweet pepper plants to foliar application of compost tea and dry yeast under soilless conditions. *Bull. Natl. Res. Cent.* **2021**, *45*, 119. [CrossRef]
92. Villecco, D.; Pane, C.; Ronga, D.; Zaccardelli, M. Enhancing sustainability of tomato, pepper and melon nursery production systems by using compost tea spray applications. *Agronomy* **2020**, *10*, 1336. [CrossRef]
93. Pane, C.; Palese, A.M.; Celano, G.; Zaccardelli, M. Effects of compost tea treatments on productivity of lettuce and kohlrabi systems under organic cropping management. *Ital. J. Agron.* **2014**, *9*, 153–156. [CrossRef]
94. González-Hernández, A.I.; Pérez-Sánchez, R.; Gómez-Sánchez, M.Á.; Morales-Corts, M.R. Compost tea as biostimulant: Promoting tomato root development. *Chem. Proc.* **2022**, *10*, 57. [CrossRef]
95. Abubaker, S.; Qrunfleh, I.; Shatnawi, M.; Ammari, T.G.; Hasan, H.; Al-Tawaha, A.R.M. The effect of compost tea on some growth and yield parameters and soil chemical properties of greenhouse tomato (*Solanum lycopersicum* L.). *Ecol. Eng. Environ. Technol.* **2024**, *25*, 362–370. [CrossRef]
96. Girshe, L.; Mengistu, H.; Tadesse, T. The effects of rate and method of aerated compost tea application on yield and yield component of tomato (*Lycopersicon esculentum* Mill.) at Burusa, South Western Ethiopia. *Int. J. Multidiscip. Curr. Res.* **2018**, *6*, 52–58. [CrossRef]
97. De Corato, U. Compost and Compost Tea from On-Farm Composted Agro-Wastes Improve the Sustainability of Horticultural Organic Cropping Systems. In *Agri-Based Bioeconomy*; CRC Press: Boca Raton, FL, USA, 2021; pp. 143–162.
98. Formisano, L.; Miras-Moreno, B.; Ciriello, M.; Zhang, L.; De Pascale, S.; Lucini, L.; Rouphael, Y. Between light and shading: Morphological, biochemical, and metabolomics insights into the influence of blue photoselective shading on vegetable seedlings. *Front. Plant Sci.* **2022**, *13*, 890830. [CrossRef]
99. Carrascosa, A.; Pascual, J.A.; López-García, Á.; Romo-Vaquero, M.; De Santiago, A.; Ros, M.; Petropoulos, S.A.; Alguacil, M.D.M. Effects of inorganic and compost tea fertilizers application on the taxonomic and functional microbial diversity of the purslane rhizosphere. *Front. Plant Sci.* **2023**, *14*, 1159823. [CrossRef] [PubMed]
100. Luo, T.; Min, T.; Ru, S.; Li, J. Response of cotton root growth and rhizosphere soil bacterial communities to the application of acid compost tea in calcareous soil. *Appl. Soil Ecol.* **2022**, *177*, 104523. [CrossRef]
101. Lazcano, C.; Domínguez, J. The use of vermicompost in sustainable agriculture: Impact on plant growth and soil fertility. *Soil Nutr.* **2011**, *10*, 187.
102. Atiyeh, R.; Lee, S.; Edwards, C.; Arancon, N.; Metzger, J. The influence of humic acids derived from earthworm-processed organic wastes on plant growth. *Bioresour. Technol.* **2002**, *84*, 7–14. [CrossRef] [PubMed]
103. Canellas, L.P.; Olivares, F.L.; Aguiar, N.O.; Jones, D.L.; Nebbioso, A.; Mazzei, P.; Piccolo, A. Humic and fulvic acids as biostimulants in horticulture. *Sci. Hortic.* **2015**, *196*, 15–27. [CrossRef]
104. Jindo, K.; Olivares, F.L.; Malcher, D.J.d.P.; Sánchez-Monedero, M.A.; Kempenaar, C.; Canellas, L.P. From lab to field: Role of humic substances under open-field and greenhouse conditions as biostimulant and biocontrol agent. *Front. Plant Sci.* **2020**, *11*, 426. [CrossRef]
105. Ali, O.A. Role of humic substances and compost tea in improvement of endogenous hormones content, flowering and yield and its components of faba bean (*Vicia faba* L.). *Ann. Agric. Sci. Moshtohor* **2015**, *53*, 373–384.
106. El-Din, A.A.E.; Hendawy, S. Effect of dry yeast and compost tea on growth and oil content of Borago officinalis plant. *Res. J. Agric. Biol. Sci.* **2010**, *6*, 424–430.
107. Li, W.; Liu, Y.; Chai, X.; He, J.; Liu, C.; Li, J. Advantages of compost tea: Promotion of nitrogen influx into the fruit and improvement of fruit nitrogen metabolism in tomato. *Plant Physiol. Biochem.* **2024**, *216*, 109184. [CrossRef]
108. Xu, D.; Raza, W.; Yu, G.; Zhao, Q.; Shen, Q.; Huang, Q. Phytotoxicity analysis of extracts from compost and their ability to inhibit soil-borne pathogenic fungi and reduce root-knot nematodes. *World J. Microbiol. Biotechnol.* **2012**, *28*, 1193–1201. [CrossRef]
109. Xu, D.-B.; Wang, Q.-J.; Wu, Y.-C.; Yu, G.-H.; Shen, Q.-R.; Huang, Q.-W. Humic-like substances from different compost extracts could significantly promote cucumber growth. *Pedosphere* **2012**, *22*, 815–824. [CrossRef]
110. Eudoxie, G.; Grogan, K.; Beckford, M.; Martin, M. Compost tea influence on lettuce (*Lactuca sativa* L.) root architecture. In Proceedings of the International Symposium on Growing Media, Soilless Cultivation, and Compost Utilization in Horticulture 1266, Portland, OR, USA, 20–25 August 2017; pp. 79–88.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.